

Simulation Inputs

IMU

- Gyroscope
- Accelerometer

Odometry

- Wheel Odometry

Initial State

- Position/Velocity
- Attitude
- EKF State
- Covariance



B Trajectories
× N Steps

§4.2 Kernel Fusion

Full-Loop Fusion

Fuse complete process
into single kernel

Register-Resident

Keep states on chip
across timesteps

Micro-Matrix Opt

Use padded layouts
and analytic inversion

§4.3 Component-Aware Mix Precision

Component-Aware

Profile precision needs
by component

Compile Specialization

Select precision
paths with constexpr

Error-Bounded Selection

Maximize speed under
accuracy constraints

§4.4 Multi-Trajectory Parallelism

Trajectory Batching

Run independent
trajectories in parallel

Batch Mapping

Assign one CUDA
block per trajectory

Register-Aware Scaling

Scale batch within
occupancy limits